RESIDUAL STRUCTURE DOES NOT ACCOUNT FOR THE RAW MODULAR EXCESS IN ZETA-ZERO GAPS

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Remark 0.1. This notebook is exploratory in nature. Its purpose is to record and organise numerical experiments on nearest-neighbour gaps between nontrivial zeros of the Riemann zeta function, with a particular focus on two aspects: the residuals of a smooth reference model and the modular distribution of raw gaps. Several sections contain formal manipulations, heuristic arguments, and speculative interpretations. Whenever a statement is rigorous it will be explicitly identified as such. All remaining claims should be regarded as tentative, heuristic, or conjectural.

1. Objects and notation

Let

$$\rho_k = \tfrac{1}{2} + i\gamma_k \tag{1.1}$$

denote the k-th nontrivial zero of the Riemann zeta function on the critical line, assuming the Riemann Hypothesis for notational convenience. We write

$$Y_k = \gamma_k \tag{1.2}$$

for the ordinates and

$$G_k = Y_{k+1} - Y_k \tag{1.3}$$

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for the raw nearest-neighbour gaps.

Remark 1.1. The notation in (1.1)–(1.3) is standard and consistent with classical references such as [1, 2]. No new assumptions are introduced here.

We consider a smooth reference model for the ordinates, denoted

$$\gamma_k^{\mathrm{RVM}}, \tag{1.4}$$

and define the residuals

$$R_k = \gamma_k - \gamma_k^{\mathrm{RVM}}. \tag{1.5}$$

The corresponding residual jumps are

$$\Delta R_k = R_{k+1} - R_k. \tag{1.6}$$

Remark 1.2. The precise construction of the reference model γ_k^{RVM} in (1.4) is not needed for the present notebook. We treat it as a smooth approximation to the ordinates whose residuals R_k and jumps ΔR_k can be computed numerically.

For the modular analysis of raw gaps we consider the fractional parts

$$x_k = \{Y_{k+1} - Y_k\} \in [0, 1), \tag{1.7}$$

and study the first circular harmonic of their empirical distribution.

2. Summary of the residual/gap matrix study

This section records the main numerical outputs of the "matrix study of residuals/gaps". The underlying computations are local in k and use the first 51 residuals and 50 residual jumps.

2.1. **Basic statistics.**

The study reports the following summary statistics for the residuals and their jumps:

$$\mathbb{E}[R_k] \approx -1.3201457, \tag{2.1}$$

$$\mathrm{std}(R_k) \approx 0.66974925, \tag{2.2}$$

$$\mathrm{std}(\Delta R_k) \approx 0.87518461. \tag{2.3}$$

Experiment 2.1. The quantities in (2.1)–(2.3) are computed from the first 51 residuals R_k and the first 50 jumps ΔR_k. Empirically one finds a negative mean residual and standard deviations of order 1 for both R_k and ΔR_k.

2.2. **Feature correlation matrix.**

The matrix study considers five features, which we denote symbolically by

$$(dG, dRVM, dR, SG, SR), \tag{2.4}$$

and reports the empirical correlation matrix

$$C = \begin{pmatrix} 1.00 & 0.69 & 0.82 & 0.76 & 0.76 \\ 0.69 & 1.00 & 0.15 & 0.08 & 0.07 \\ 0.82 & 0.15 & 1.00 & 0.98 & 0.97 \\ 0.76 & 0.08 & 0.98 & 1.00 & 1.00 \\ 0.76 & 0.07 & 0.97 & 1.00 & 1.00 \end{pmatrix}. \tag{2.5}$$

Experiment 2.2. The matrix (2.5) shows that dR is strongly correlated with SG and SR (correlations 0.98 and 0.97), while $dRVM$ has only weak correlation with dR (correlation 0.15). This indicates that the residual jumps ΔR_k are not linearly controlled by the smooth model increments $dRVM$, but they are tightly linked to the gap-related features SG and SR.

2.3. **Local PCA of features.** A principal component analysis of the five-dimensional feature vector (2.4) yields that the first two principal components explain essentially all of the variance:

$$\text{PC1 variance share} \approx 74.82\%, \tag{2.6}$$

$$\text{PC2 variance share} \approx 24.50\%, \tag{2.7}$$

$$\text{PC1+PC2 variance share} \approx 99.323\%. \tag{2.8}$$

Remark 2.3. The near saturation in (2.8) suggests that the local feature cloud is effectively two-dimensional in the PCA sense. This is a structural observation about the joint distribution of $(dG, dRVM, dR, SG, SR)$, but it does not by itself imply any simple temporal dynamics for the residuals.

2.4. **Hankel singular values of residual jumps.** The Hankel matrix built from the residual jumps ΔR_k is analysed via its singular values. The study reports that the leading singular value accounts for

$$\sigma_1 \approx 10.193\%, \tag{2.9}$$

and the first five singular values together account for

$$\sigma_1 + \cdots + \sigma_5 \approx 45.730\%. \tag{2.10}$$

In logarithmic scale, the singular values are given (for the first few) by

$$\log_{10}(\sigma_j) \approx 1.15,\ 0.417,\ -0.316,\ -1.05,\ -1.78,\ -2.52,\ \ldots \tag{2.11}$$

for $j = 1, 2, \ldots$.

Experiment 2.4. The Hankel singular spectrum of ΔR_k does not collapse to rank one: the leading singular value carries only about 10% of the total weight, and the first five together less than 50%. This indicates that the residual jumps do not exhibit a dominant low-rank temporal structure.

We now include a PGFPlots figure for the logarithmic singular values reported in (2.11).

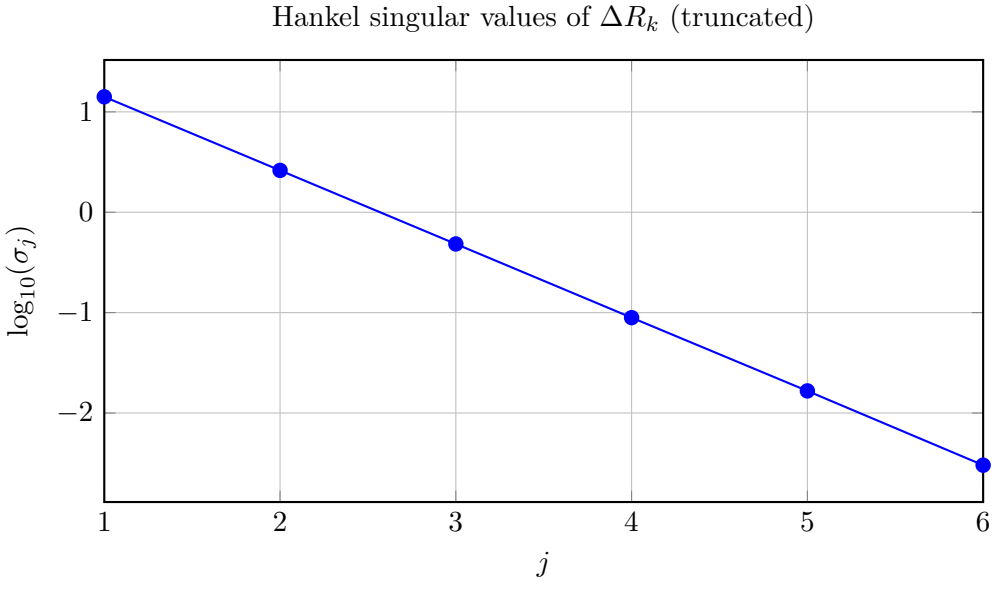


FIGURE 1. Logarithmic Hankel singular values of the residual jumps ΔR_k, using the reported values in (2.11).

2.5. **Sliding-window PCA of residuals.** A sliding-window PCA of the residual sequence R_k produces eigenvalues whose logarithms are reported as

$$\log_{10}(\lambda_j) \approx 0.228,\ -0.63,\ -1.49,\ -2.34,\ -3.2,\ -4.06,\ \dots \tag{2.12}$$

for $j = 1, 2, \dots$. The variance shares are summarised as

$$\text{PC1 variance share} \approx 15.00\%, \tag{2.13}$$

$$\text{PC1+PC2 variance share} \approx 28.406\%. \tag{2.14}$$

Experiment 2.5. The sliding-window PCA of R_k shows that the leading principal components explain only a modest fraction of the local variance: about 15% for PC1 and 28% for PC1+PC2. This is consistent with a relatively diffuse local geometry for the residual sequence.

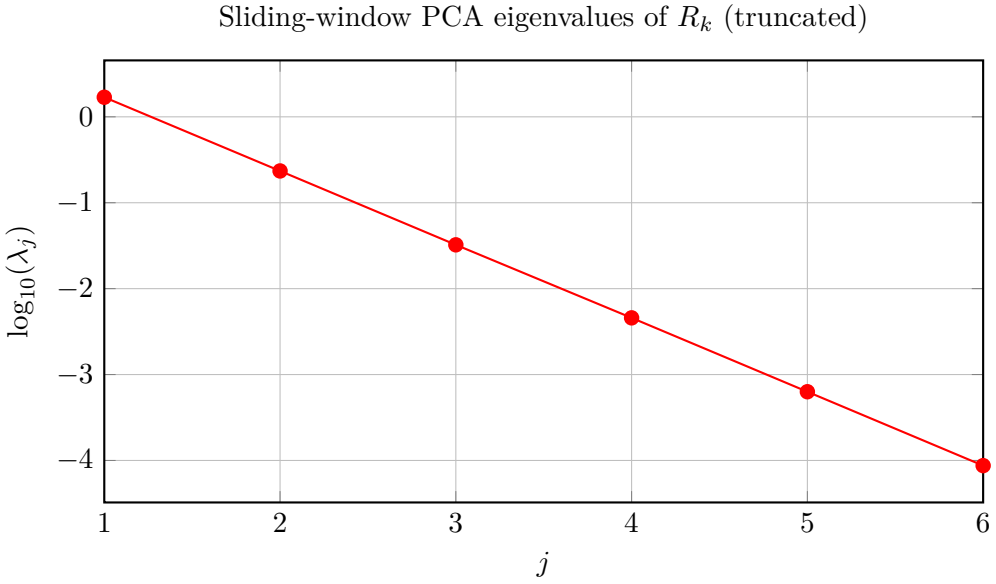


FIGURE 2. Logarithmic eigenvalues from a sliding-window PCA of the residuals R_k, using the reported values in (2.12).

3. Summary of the raw modular excess study

We now summarise the "raw modular excess" short communication, which investigates the fractional parts x_k in (1.7) for large samples of zeros.

3.1. **First harmonic statistic.** For a sample (x_k) in $[0, 1)$, the study defines

$$a = -2 \sum_k \cos(2\pi x_k), \tag{3.1}$$

$$b = 2 \sum_k \sin(2\pi x_k), \tag{3.2}$$

$$A = \sqrt{a^2 + b^2}, \tag{3.3}$$

and considers the fitted first harmonic

$$p(x) = 1 - a\cos(2\pi x) + b\sin(2\pi x), \tag{3.4}$$

together with the phase of the maximum

$$x_{\max} = \frac{1}{2\pi} \arg(-a + ib) \pmod 1. \tag{3.5}$$

Remark 3.1. The use of the first circular harmonic (3.1)–(3.5) is standard in the analysis of angular or modular data. Here it is applied to the raw gap fractional parts x_k.

3.2. **Null models.** Two scale-preserving null models are considered. Let h_k denote a local scale factor depending on Y_{k+1}. The first null is a local GUE-based model, denoted

$$\Delta Y_k^{\mathrm{GUE}} = h_k^{\mathrm{GUE}}, \tag{3.6}$$

while the second null is a shuffled-real model that preserves the empirical unfolded spacing distribution but destroys temporal order:

$$\Delta Y_k^{\mathrm{shuf}} = h_k^{\mathrm{shuf}}. \tag{3.7}$$

Remark 3.2. The GUE-scale null (3.6) tests the agreement of the raw modular distribution with a local random-matrix prediction. The shuffled-real null (3.7) is stricter: it keeps the empirical unfolded spacing distribution and tests whether any remaining modular excess requires temporal coupling to the slowly varying scale h_k.

3.3. **Numerical results for the first harmonic.** For $N = 1{,}000{,}000$ and $N = 2{,}000{,}000$ zeros, the study reports the first-harmonic coefficient a for the real gaps, the GUE-scale null, and the shuffled-real null. The results are summarised in Table 1.

TABLE 1. First harmonic of raw gap fractional parts. Z-scores compare the real value against 50 null draws.

N	a_{real}	a_{GUE}	a_{shuf}	Z_{GUE}	Z_{shuf}	$x_{\max}$
1,000,000	0.616383	0.573036	0.618266	30.25	−3.70	0.544853
2,000,000	0.710876	0.664228	0.712014	49.22	−3.79	0.519464

For $N = 2{,}000{,}000$ the relative excess over the GUE-scale null is

$$\frac{a_{\mathrm{real}} - a_{\mathrm{GUE}}}{a_{\mathrm{GUE}}} \approx 7.02\%. \tag{3.8}$$

The shuffled-real null gives

$$\frac{a_{\mathrm{real}} - a_{\mathrm{shuf}}}{a_{\mathrm{shuf}}} \approx -0.16\%. \tag{3.9}$$

Experiment 3.3. At $N = 2{,}000{,}000$ zeros, the real raw gaps exhibit a first-harmonic amplitude a_{real} about 7.02% larger than the GUE-scale null a_{GUE}, as in (3.8). However, once the empirical unfolded spacings are shuffled and remultiplied by the same local scale h_k, the shuffled-real null a_{shuf} nearly matches a_{real}, with a residual difference of about -0.16% as in (3.9).

We include a PGFPlots figure comparing the three a-values for $N = 2{,}000{,}000$.

3.4. **Location of the modular peak.** The study reports that the raw gap fractional parts are more concentrated near

$$x \approx 0.51946 \tag{3.10}$$

than predicted by the local GUE scaling. The absolute ordinates themselves pass a sanity check: for $N = 2{,}000{,}000$ the first harmonic amplitude of $\log Y_k \bmod 1$ is only about 1.535×10^{-5}, indicating that the signal is located in the increments.

Remark 3.4. The observation (3.10) indicates a genuine modular feature of the raw gaps: their fractional parts are not uniformly distributed modulo one at the level of the first harmonic. However, the shuffled-real null shows that this feature is essentially explained by the empirical unfolded spacing distribution, rather than by a strong temporal coupling to the local scale.

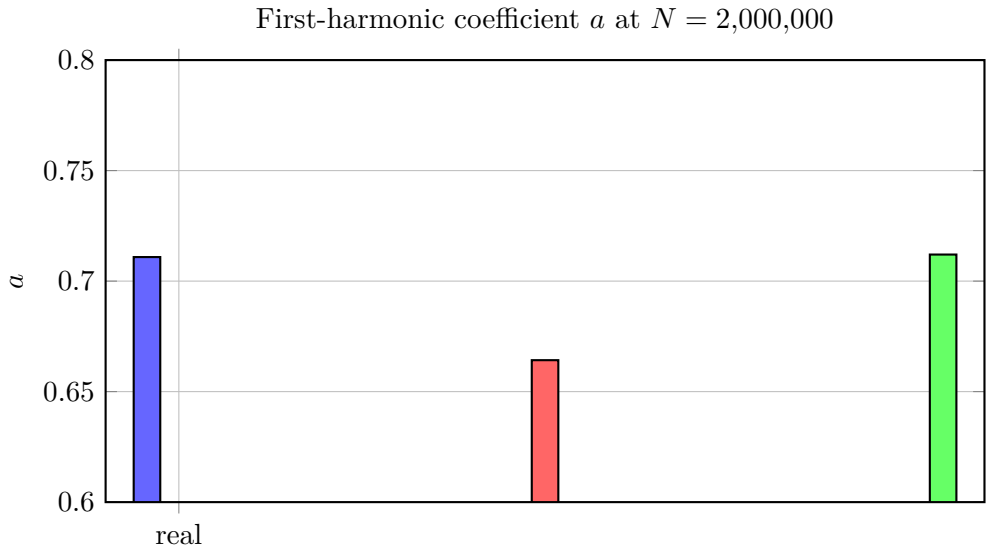


FIGURE 3. First-harmonic coefficient a for the real gaps, the GUE-scale null, and the shuffled-real null at $N = 2{,}000{,}000$, using the values in Table 1.

4. Integrated interpretation

We now combine the residual/gap matrix study (Section 2) with the raw modular excess study (Section 3).

4.1. **Residual structure versus modular excess.** The matrix study shows that:

- The feature space $(dG, dRVM, dR, SG, SR)$ is effectively two-dimensional in the PCA sense (Experiment 2.2, Remark 2.3).
- The Hankel singular spectrum of ΔR_k does not exhibit a dominant low-rank structure (Experiment 2.4).
- The sliding-window PCA of R_k has relatively diffuse leading components (Experiment 2.5).

In particular, there is no evidence of a simple low-rank temporal dynamics for the residuals.

On the other hand, the modular excess study shows that:

- The raw gap fractional parts exhibit a clear first-harmonic excess over the GUE-scale null (Experiment 3.3).
- This excess is almost entirely absorbed by the shuffled-real null, which preserves the empirical unfolded spacing distribution but destroys temporal order (Equations (3.8) and (3.9)).

Remark 4.1. Taken together, the two studies indicate that the observed raw modular excess in zeta-zero gaps is not explained by a simple low-rank or linearly correlated structure in the residuals R_k or their jumps ΔR_k. Instead, the excess is essentially accounted for by the empirical unfolded spacing distribution itself.

4.2. **Heuristic formulation of the main conclusion.** We now state the main conclusion in heuristic form.

Heuristic 4.2. Let $G_k = Y_{k+1} - Y_k$ be the raw gaps and $x_k = \{G_k\}$ their fractional parts. Let R_k be the residuals of a smooth reference model and ΔR_k their jumps. The numerical evidence suggests that:

(a) The sequence (R_k) and its jumps (ΔR_k) do not exhibit a dominant low-rank temporal structure, as measured by Hankel singular values and sliding-window PCA.

(b) The first-harmonic excess of the modular distribution of (x_k) over a local GUE-scale null is real but small (about 7% at $N = 2{,}000{,}000$ zeros).
(c) This excess is almost entirely reproduced by a shuffled-real null that preserves the empirical unfolded spacing distribution while destroying temporal order.

Heuristically, this indicates that the raw modular excess is traceable to the empirical unfolded spacing law and not to a strong temporal coupling between the residuals and the local scale.

Warning 4.3. Heuristic 4.2 is based on finite-range numerical experiments and on specific choices of null models. It should not be interpreted as a rigorous theorem about the asymptotic behaviour of zeta-zero gaps.

5. Open questions and future directions

We conclude with a few questions suggested by the experiments.

Question 5.1. Can one construct a refined reference model for the ordinates γ_k whose residuals R_k have provably small Hankel rank or controlled temporal structure, and would such a model change the modular excess picture?

Question 5.2. Is there a theoretical framework that directly links the empirical unfolded spacing distribution of zeta zeros to the observed first-harmonic modular excess, beyond random-matrix heuristics?

Remark 5.3. Future work may include: more extensive Hankel and PCA analyses at higher heights, alternative null models for modular statistics, and a closer comparison with rigorous results on pair correlation and higher-order spacing statistics (see for instance [3, 4]).

References

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